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Modelling the climate change and cotton yield relationship in Mississippi: Autoregressive distributed lag approach

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ABSTRACT

Development of mitigation strategies to combat climate change necessitates an advanced analysis of the historical connection between crops and climate. Such an analysis is lacking for the cotton (*Gossypium hirsutum* L.)-climate research in Mississippi (MS). Hitherto, research has been confined to small-scale experimental settings, leaving an opportunity to explore large-scale inferences. Therefore, the present study aimed to compute MS climatic trends during the cotton growing period (CGP) from 1970 to 2020 using the Mann-Kendall and Sen slope methods. The impact of climate change on MS cotton yield was assessed using the autoregressive distributed lag (ARDL) model. The climatic variables considered were maximum temperature (Tmax), minimum temperature (Tmin), diurnal temperature range (DTR), precipitation (PR), and CO_2 emissions (COE). A required series of statistical tests, including pre- and post-analysis, model robustness, and goodness-of-fit were performed, and data met all criteria. Results revealed that Tmin (79.6 %) contributed more than Tmax (20.4 %) to the MS-climate warming over CGP. From 1970 to 2020, the Tmax, Tmin, DTR, and PR changed by + 0.30 °C, +1.17 °C, −1.07 °C, and + 22.54 mm, respectively, exhibiting change rate per decade of + 0.06 °C, +0.23 °C, −0.21 °C, and + 4.42 mm, respectively. Precipitation had no effect on cotton yield in the long or short-term. However, cotton yield significantly decreased with a rise in Tmax, and increased with a rise in Tmin and COE in the long-term. Conclusively, a 1 °C increase in Tmax reduced cotton yield by 6.1 %, a 1 °C increase in Tmin improved it by 5.5 %, and a unit increase in COE increased it by 0.45 % over the long run. Overall, the crop-climate link in MS cotton marked a varied sensitivity towards short and long-term, indicating the need to reassess current mitigation strategies. Additionally, testing the best agronomic practices in a controlled environment at the actual rates of climate change identified by the current study could provide cotton stakeholders with more precise and valuable insights.

1. Introduction

The rising rate of greenhouse gas emissions, mainly anthropogenic carbon emissions, has already put our ecosystem on a path towards severe and irrevocable global warming (Solomon et al., 2009). This pathway drives drought, elevated temperatures, snowstorms, and floods with greater frequency and heightened intensity (Gathen et al., 2021). Nine of the ten warmest years since 1860 were after 1990, with 1998 and 2001 ranked the warmest years in the past 1000 years (Albritton et al., 2001; O'Neal et al., 2005). The current generation of agriculturists is first in line to experience the effects of anthropogenic climate change (Brown, 2019). Agriculture as we know it today has evolved over eleven thousand years in a relatively stable climate allowing production output maximization (Brown, 2013). However, the climate is now abruptly shifting and consequently desynchronizing the agricultural systems (Naylor, 2021; Ranguwal et al., 2023). Within the US, Mississippi (MS), a southern state, is already threatened by hurricanes, severe flooding, heatwaves, tornadoes, and prolonged drought, which are anticipated to worsen in the future (EPA, 2016). Mississippi is the 3rd wettest state in the US with 1420 mm (55.9 in.) of precipitation (PR) annually (NOAA, 2021). Recently, 2016–2020 marked MS's warmest 5-year span on record (NOAA, 2021). By the year 2100, temperatures are predicted to

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surpass 35 °C (95°F) on 30 to 60 days year^{-1}, compared to 15 days year^{-1} in MS presently (EPA, 2016). Additionally, MS has a significant impact on the agroeconomic setting in the US, covering 4.21 million hectares under agriculture, generating $8.33 billion revenue, and engaging 17.4 % of its population in agriculture (MDAC, 2021).

Mississippi is one of the "cotton (*Gossypium hirsutum* L.) belt" states of the US, accounting for 7.06 % of cropland and yielding 8.04 % of the national production (USDA-NASS, 2023; WA, 2023). Annually, MS grows cotton on 210 thousand hectares, producing 1.18 million 480-pound bales, generating $558 million. In the global context, cotton is the primary fiber crop, grown by more than 100 nations (Aslam et al., 2023; Tariq et al., 2023). Being a prominent cash crop, cotton is used as a raw ingredient in the garment industry with an annual total global revenue generation of $600 billion (Khan et al., 2020). For the US, cotton generates $25 billion annually through various products such as yarns, cotton-seed oil, livestock feeding, and fabrics (Raper et al., 2019; Adeleke, 2023).

Moreover, cotton is a crucial crop to research from a climatic standpoint due to its low environmental sensitivity to climatic changes (Amin et al., 2017). Cotton can resist temperature and drought stress due to its vertical tap root system (Ton, 2012), which suits temperate, tropical, and subtropical climates (Dang et al., 2023). Besides, temperature affects the rates of fruit development, photosynthetic activity, respiration, and evaporation rates during the cotton growth period (CGP) (Turner et al., 1986). Depending on warming rates, temperature may also intensify water stress, negatively affecting the growth of cellulose cells in cotton bolls, and lowering the fiber quality (Azhar and Rehman, 2018). Alternatively, increased temperatures are beneficial at both the start and end of the CGP, but detrimental amid the CGP for cotton yield (Bange, 2007). However, because cotton is a C3 crop, it benefits more than C4 crops from the fertilization effects caused by increased CO_2 emissions (COE) (Reddy et al., 2002). Therefore, in terms of altering cotton production, climate change is anticipated to have diversified sensitivity to climatic conditions, spatial variations, and regional differences (Li et al., 2021). Investigations at a finer level (statewide) are thus required to discover these sensitivity levels to comprehend better the crop-climate nexus (Schlenker and Roberts, 2009). These finer-scale investigations are crucial in identifying mitigation strategies and improving resilience to climatic effects, aiding in better decision-making (Williams et al., 2015).

Nonetheless, in MS, most of the climate-cotton research to date has been done in experimental settings. However, studies exploring it on a larger scale incorporating actual data from production farms are scarce (Reddy et al., 2002; Anapalli et al., 2016). Consequently, the rate and magnitude of climate changes during CGP and its impact on cotton production at the MS scale are unknown. Therefore, the current study employed the advanced ARDL model to assess the crop-climate link in cotton to answer the following objectives:

i) Compute the maximum temperature (Tmax), minimum temperature (Tmin), diurnal temperature range (DTR), and PR trends during CGP in MS.
ii) Quantify the impact of climate change on cotton yield in MS.

2. Methodology

2.1. Data

The study utilized the previous 50 years' (1970–2020) time series data for MS (Fig. 2) with data description mentioned in Table 1.

Time series data, and the data sources used are regarded as reliable and have been extensively used in several crop-climate studies (Daly and Bryant, 2013; Chandio et al., 2022). The data sources utilized high-resolution (4–5 km) specifications for transcoding the data units from grid observatory to achieve adequate spatial sampling (NOAA, 2021). The NOAA gathers data from a broad range of weather station networks worldwide (40,000) (Van Wart et al., 2013). The collected data is processed using various quality control procedures to account for altitude/topographical differences in weather stations and the historical bias adjustments to ensure data integrity (Vose et al., 2014). Three categories including high variability, mid-level variability, and low variability in altitude/topographical variations exists in the US. Moreover, MS lies among the low variability states with the least topographical diversity and altitude changes, along with other states such as Florida, Delaware, and Louisiana (McGuire et al., 2020). However, for all regions, NOAA strictly follows its robust computational methods to process station's raw data (Vose et al., 2014). Furthermore, readers interest in an in-depth information regarding NOAA's uses of computational methods are directed to Vose et al. (2014), briefly, Vose et al. (2014) delineates the following:

Table 1
Variables, data sources, and description of the calculative work performed on the data prior to feed in the study model, as used in this study.

Variables	Data source	Calculative work on data before inputting in the study model
Tmax (°C)	NOAA https://www.nass.usda.gov/	Tmax, Tmin, and DTR were averaged over CGP for computing growing season trends (Question 1) and for crop-climate impact estimation by the ARDL model (Question 2)
Tmin (°C)	NOAA https://www.nass.usda.gov/	
DTR (°C)	NOAA https://www.nass.usda.gov/	
PR (mm)	NOAA https://www.nass.usda.gov/	Precipitation was totalled over CGP for computing growing season trends (Question 1) and overall crop-climate impact by ARDL model (Question 2)
COE (Mmt)	USEIA https://www.eia.gov/	For COE, yearly averages were used for Question 2. Growing season trend (Question 1) was impossible as monthly data was unavailable.
Cotton yield (Mg ha^{-1})	USDA-NASS https://www.nass.usda.gov/	No additional calculative work was needed.
HA (ha)	USDA-NASS https://www.nass.usda.gov/	No additional calculative work was needed.

NOAA is the National Oceanic and Atmospheric Administration, USDA-NASS is the United States Department of Agriculture's National Agricultural Statistics Service, USEIA is the United States Energy Information Administration, and PRISM is the parameter elevation regressions on independent slopes model. Tmax is maximum temperature, Tmin is minimum temperature, DTR is diurnal temperature range, PR is precipitation, COE is CO2 emissions, °C is degree Celsius, mm is millimeter, Mmt is million metric tons, Mg/ha is megagram per hectare, and ha is hectare.

a. Altitude adjustment in climatological interpolation: A 3-step grid approach is applied to interpolate the raw station-data.
b. Step-1, the climate normal (average of base-period *i.e.*,1981–2010) per station is calculated. In case, any station has inadequate base-period data, 30-year average is utilized, and modifications are applied to align with the original base-period (1981–2010). Then, the climate normal are adjusted following the relationship between climatic parameters and altitude, to account for differences in topography, using equation (1) as follows.

$$C(x,y,z) = C_{normal}(x,y) + \chi(z - z_{mean}) \quad (1)$$

where, $C(x,y,z)$ denotes adjusted value of climate normal for station location (*x, y*) at altitude (*z*). $C_{normal}(x,y)$ is the baseline climate normal for station location (*x, y*). χ is the lapse rate, specifying the rate at which the climate parameter changes with altitude. z is altitude at station location (*x, y*) and z_{mean} is the reference altitude for the normal.

c. Step 2, the adjusted values of climate normal are interpolated on a high-density grid using a thin-plate smoothing spline (TPSS) method (Hutchinson, 1995). The TPSS technique generates normal grids by fitting smooth/continues surfaces to the climatic datapoints considering complex geographical/altitude variations across large regions (Hutchinson, 1995). The TPSS technique is robust in ensuring smooth interpolation by minimizing the sum of squared differences of measured datapoints from the spline function. The TPSS method works as per equation (2) which is as follows:

$$D(x,y) = \sum_{j=1}^{T} \gamma_j \psi\left(\sqrt{(x-x_j)^2 + (y-y_j)^2}\right) \tag{2}$$

where, $D(x, y)$ is function of x and y. Summation denotes the sum up to T terms. γ_j denote coefficients tied to j^{th} (or each) term. $\psi(r)$ represents radical function having the form of $\psi(r) = r^2 \log(r)$ for thin plate spline. $\sqrt{(x-x_j)^2 + (y-y_j)^2}$ signifies the Euclidean distance of point (x, y) from every (x_j, y_j) pair.

d. Step 3, following equation (3), the climatic anomaly values are merged with the adjusted climatic normal for generating final climate values.

$$E_{final}(x,y,z) = C(x,y,z) + D(x,y) \tag{3}$$

Where, $E_{final}(x, y, z)$ is the final climate data. $C(x, y, z)$ denotes the adjusted climatic normal for a station located at an altitude of z. $D(x, y)$ is the interpolated climatic anomaly of a station located as coordinates (x, y).

e. Area-weighted average (Sharples et al., 2005): The interpolated climate values are then averaged (area-weighted) per climate division, confirming that all stations are weighted proportional to their area coverage. Moreover, this decreases the biasness caused by unequal allocation of stations and ascertains the proportional representation of each area's climatic attributes in the conclusive averages. The conclusive averages here represent climate values for each county of MS.

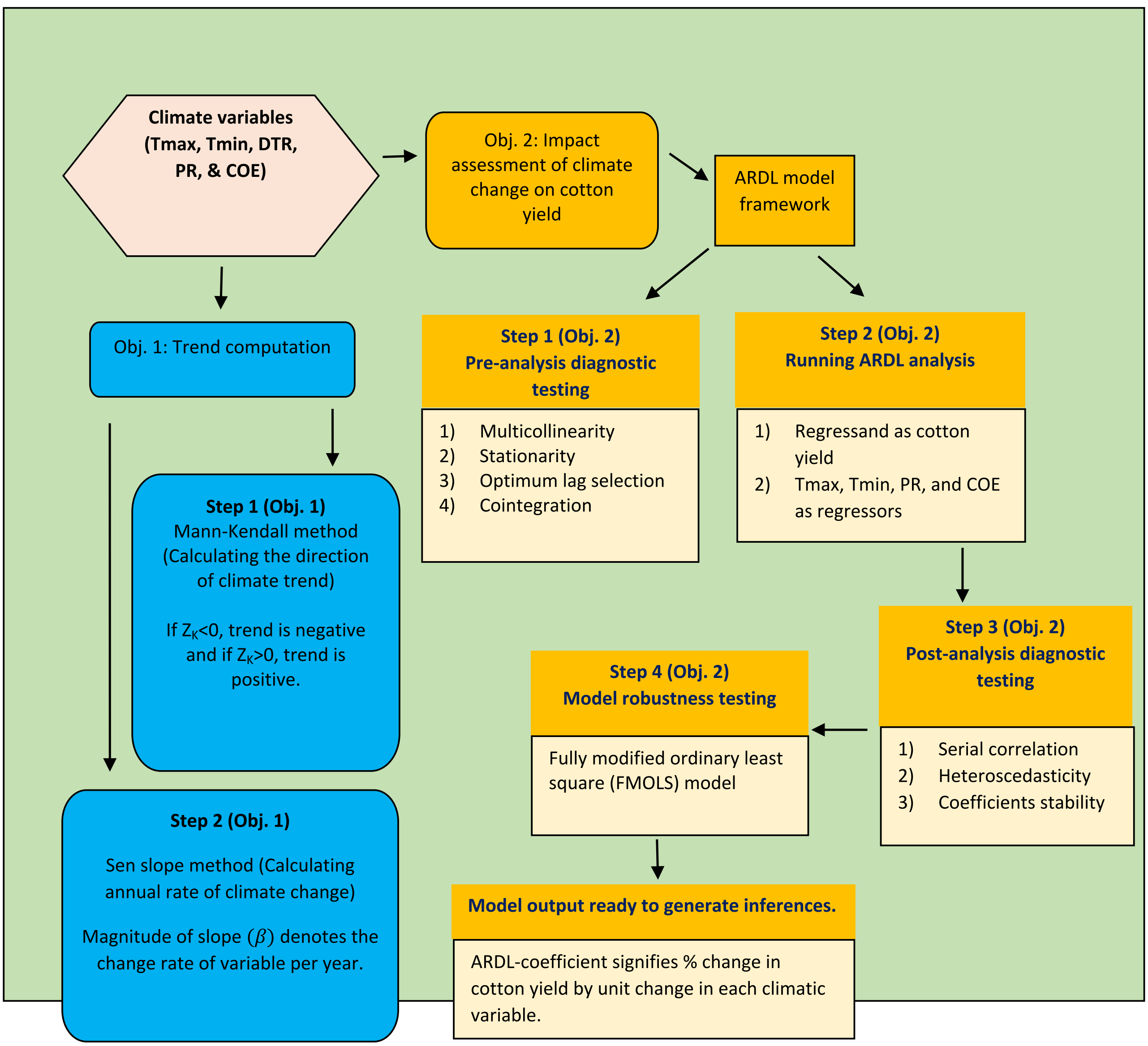

Fig. 1. A flowchart showing each workflow step in the methodology for the two study objectives. The first objective, estimating the trend for each of the four climatic variables—maximum temperature (Tmax), minimum temperature (Tmin), diurnal temperature range (DTR), and precipitation (PR)—is shown on the left in blue boxes. The second objective, quantifying the overall effect of change in climatic variables on cotton yield, is shown on the right in yellow boxes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

f. Missing Data: In case of missing data at a station, NOAA applies regression (least absolute deviation) to determine incomplete records using adjacent stations' data, with similarity observed by an agreement index (Mielke and Berry, 2007).
g. Bias adjustments: Historical bias (if or wherever needed) adjustments are performed using pairwise approach and corrections are made as per either the documented or undocumented changes in observation time, instrumentation, and station location to affirm that temporal trends precisely reflect actual climate signals instead of getting biased by data collection errors (Menne and Williams, 2009).
h. Quality assurance: Lastly, all the records are periodically scrutinized running a series of logical, serialized, and geospatial quality checks to identify inaccurate values (if exist) confirming data integrity (Durre et al., 2010).

Because of the strengths of NOAA's rigorous methodology (points i-v) in constructing regional time series for climate data, it has been widely used as data source for climate related research (Preisler and Westerling 2007; McCabe et al. 2007; Goodrich 2007; Rogers et al. 2007; Goodrich and Ellis 2008; Grundstein 2009; Mauget et al. 2009; Seager et al. 2009; Myoung and Nielsen-Gammon 2010; Anderson et al. 2010; Van Wart et al., 2013; Konduri et al., 2020; Da et al., 2022).

In the current study, the independent variables were Tmin, Tmax, PR, DTR, and COE while harvested area (HA) and cotton yield were taken as control input and dependent variables, respectively (Fig. 1). The cotton-growing counties' area-weighted averages of Tmax, Tmin, DTR, and PR served as a single geographical unit for MS, resulting in a time-series for each variable that spanned 50 years. Data on COE was accessible only on an MS scale (Table 1). The CGP was taken as April-August following USDA calendar guidelines. As suggested in previous studies, additional calculations to prepare data for modeling are presented in Table 1 (Agbenyo et al., 2022; Pickson et al., 2020).

2.2. Climatic trend analysis (yr. 1970–2020)

For computing the trend in Tmax, Tmin, DTR, and PR, the non-parametric test Mann-Kendall (Mann, 1945; Kendall, 1948) and Sen slope (Sen, 1968) were used (Fig. 1). These methods are computationally simple, robust against missing or below threshold values (if any), and capable of handling non normal data (Sanogo et al., 2023). Moreover, the World Meteorological Organization (WMO, 2018) approved these trend detection methods. The Mann-Kendall approach compares values from a specific time range using relative ordering to generate Kendall

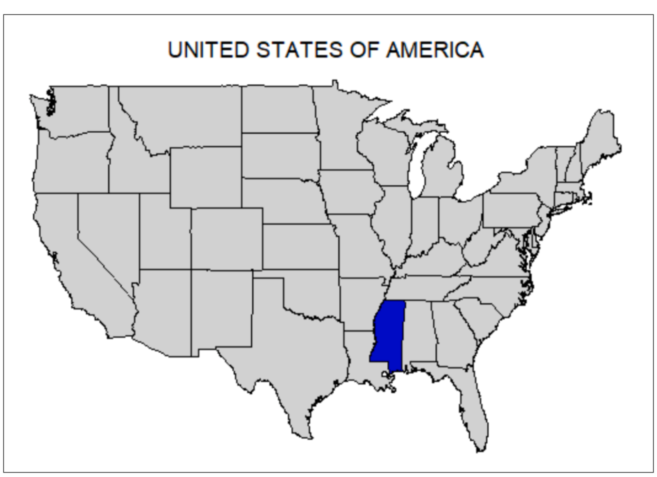

Fig. 2. The Mississippi (study area) is highlighted in blue on the United States map. The 'maps' R package was used to create the map (https://cran.r-project.org/package=maps). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

statistics to assess whether a variable is increasing, decreasing, or remaining stable (Sanogo et al., 2023). The fundamental formula for the Mann-Kendall test is as follows:

$$\text{Kendall statistics (S)} = \sum_{p=1}^{r-1} \sum_{q=p+1}^{r} \text{signum}(X_q - X_p) \tag{4}$$

The mathematical function signum in equation (4) can have one of three values (+1, 0, or −1), depending on whether Xq − Xp is greater than, equal to, or less than zero. "Xp" and "Xq" are the successive data values for years "p" and "q" in the time series, and "r" denotes the number of datapoints. If r is less than or equal to 10, Kendall statistics (S) equation-1 defines the trend; however, if r is greater than 10, as in the current study, Kendall test standardized value (Z_k) from equation-6, which is derived from S using equations (5) and (6), decides the trend (Sanogo et al., 2023).

$$\text{Variance (S)} = \frac{r(r-1)(2r+1) - \sum_{p=1}^{n} t_p(t_p-1)(2t_p+5)}{18} \tag{5}$$

In equation (5), "tp" is data point numbers in the pth group, and "n" is for the tied groups, denoting data points with common values but different rank number positions.

$$Z_K = \frac{S \pm 1}{\sqrt{\text{Variance (S)}}} \tag{6}$$

Equation (6) employs S-1 if S>0 and S+1 if S<0 as well as Z_k equals to zero if S is zero. $Z_k > 0$ denotes a positive trend, while $Z_k < 0$ denotes a negative trend (Jiqin et al., 2023).

The Sen's slope calculates the magnitude and direction of the monotonic trend or rate of change in year^{-1} in climatic variables (Fig. 1). Computing the rate of change helps determine the rate at which ecosystems must rebalance to achieve long-term equilibrium (Smith et al., 2015). Following is the equation for the Sen slope method:

$$\beta = median\left[\frac{(x_q - x_p)}{(q-p)}\right] \tag{7}$$

Equation (7)'s terms x_q and x_p are the study variables' time series at the qth and pth years, respectively. For all $p < q$, values of $\beta > 0$ indicate increasing, while $\beta < 0$ imply a decreasing trend. Additionally, the pre-Whitening procedure was used as guided by Katipoğlu (2023), before feeding the datapoints to the Mann-Kendall and Sen slope equations to ensure that the trend estimation process was free from bias caused by serial correlation. Zhang et al. (2000)'s method and Wang et al. (2001)'s method are amongst other widely used methods of pre-Whitening, suggested by the Intergovernmental Panel on Climate Change for climate trend computation. Readers interested in an in-depth understanding of the two methods, as mentioned earlier, are directed to read Gocic and Trajkovic (2013) or Mohammad et al. (2022).

2.3. Study model: Autoregressive distributive lag model (ARDL)

The ARDL model (Pesaran and Pesaran, 1997) is a time series econometric model well documented to be of use in a large body of literature evaluating the crop-climate nexus, especially when the data is available at the macro-level such as national, regional, or state level. However, the mandatory requirement for an ARDL model is continuous data (minimum 30 years) (ANL, 1994). The ARDL model excels at addressing endogeneity, small sample sizes, and performs well when different stationarity levels are found in datasets (Warsame et al., 2021). Crop-climate datasets often exhibit endogeneity, the tendency of independent variables to correlate with the error terms (Lobell and Field, 2007). The ARDL model offers additional options for choosing the optimum lag length/order for its independent and dependent variables (Pickson et al., 2020). The lag order enables the model to include lagged (previous year) values of variables, and regress them against the single dependent variable, cotton yield in this case (Agbenyo et al., 2022). By doing this, especially for a long-term consistent dataset, the signals related to the indirect climate-crop effect become stronger and easier to capture, which would otherwise be part of the residual (Pickson et al., 2020). In agriculture, there are carryover (residual) effects of the prior year's inputs on the following year's crop yield (Mello et al., 2017). For example, when slow release or granular forms are used, fertilization carryover effect may become available later and favor following year yield (Fachini et al., 2021). Similarly, late-season rainfall and the roll-over effects of cover crops are some examples of residual carryover effects (Bergkvist et al., 2011). Regressing yield or other input's lag value against the yield of the following year and enabling the lag order to determine the period (measured in years) for consideration, the ARDL model accounts for these carryover effects by incorporating them into the crop-climate linkage results (Warsame et al., 2021). Moreover, it is well-documented that climate change has direct and indirect effects on crops due to abiotic and biotic stresses (Raza et al., 2019). Specifically, the direct effects are those that the changing climate has on yield in the immediate and subsequent years, referred to as *short-term effects* (Raza et al., 2019). However, this immediate effect cumulatively transfers to permanent effects in the long run by affecting the soil processes and properties, ability to use nutrients, and other inputs, referred to as *long-term effects* (Chandio et al., 2022). The error correction model (ECM) is one of the features of the ARDL model that allows it to be autonomous in capturing these short- and long-term climatic effects simultaneously (Sharma et al., 2023a). The coefficient of ECM determines the rate (percentage year^{-1}) by which the immediate (short-run) effects transfers cumulatively to form basis for permanent (long-term) effects (Nkoro and Uko, 2016).

Beyond climatic parameters, other unobserved factors such as edaphic parameters, technological advancements, and farmer's self-regulatory adjustments, *i.e.*, planting dates, seed cultivars selection, and crop mix, affect the yield (Irmak et al., 2022). Hence, the study performed the most used technique of first-differencing the data by subtracting the data values of one year from the following year to minimize the influence of aforesaid unobserved factors (Ding and Shi, 2022). This detrending technique enabled the study to capture the pure climatic effects possible on cotton yield (Lobell and Field, 2007).

The ARDL model equation (8), adopted from similar previous studies (Pickson et al., 2020, Fig. 1), is used here as follow:

$$\begin{aligned}\Delta lnY_{it} = \sigma_0 &+ \sum_{j=1}^{n}\gamma_1\Delta\ln(Y)_{t-j} + \sum_{j=1}^{n}\gamma_2\Delta ln(Tmax)_{t-j} + \sum_{j=1}^{n}\gamma_3\Delta ln(Tmin)_{t-j} \\ &+ \sum_{j=1}^{n}\gamma_4\Delta ln(PR)_{t-j} + \sum_{j=1}^{n}\gamma_5\Delta\ln(CO_2)_{t-j} + \sum_{j=1}^{n}\gamma_6\Delta ln(HA)_{t-j} \\ &+ \sum_{j=1}^{n}\alpha_1\Delta ln(Y)_{t-j} + \sum_{j=1}^{n}\alpha_2\Delta\ln(Tmax)_{t-j} + \sum_{j=1}^{n}\alpha_3\Delta ln(Tmin)_{t-j} \\ &+ \sum_{j=1}^{n}\alpha_4\Delta ln(PR)_{t-j} + \sum_{j=1}^{n}\alpha_5\Delta ln(CO_2)_{t-j} + \sum_{j=1}^{n}\alpha_6\Delta ln(HA)_{t-j} \\ &+ \varphi(ECT)_{t-j} + \varepsilon_t\end{aligned} \tag{8}$$

$\boldsymbol{Y}$ denotes cotton yield, $\boldsymbol{\sigma_0}$ is the intercept, $\boldsymbol{j}$ is the lag order with $\boldsymbol{n}$ as the maximum lag length, Δ represents the first differencing, $\boldsymbol{\varepsilon_t}$ is the error term, $\boldsymbol{t}$ is the time in year, γ_1 **to** γ_6 represent coefficients of long-term climatic effects on Y, $\boldsymbol{\alpha_1}$ **to** $\boldsymbol{\alpha_6}$ are coefficients of short-term effects, ECT is the error correction term and $\boldsymbol{\varphi}$ is ETC's coefficient that estimates the rate at which short-term climatic impacts transfers cumulatively to form a basis for long-term permanent effects.

2.4. The ARDL calculative framework

The ARDL calculative framework requires an integral set of statistical steps that must be followed to compute the relationship between the causal variables and the response variable. This includes test for unit root or stationarity, multi-collinearity evaluation, ideal lag length criteria computation, co-integration assessment, subsequent post-analysis testing, and sensitivity or robustness verification of the ARDL model's output.

2.4.1. Unit root/non-stationarity testing

A unit root problem arises when data is not stationary over the course of the study, meaning that the mean, covariances, and variances are function of time or non-constant *i.e.*, diverging/converging with time (Lanne et al., 2002). Non-stationarity can arise for various reasons such as breaks or deterministic/stochastic trends in the data, making it difficult for a single regression coefficient to adequately capture the true nature of the dynamic relationship between the variables (Schaffer et al., 2021). This could lead to spurious regressions, especially when working with long-term and time-series datasets (Lanne et al., 2002). Hence, the literature suggests detrending the data, first differencing, second differencing, or logarithmic transformations to minimize the non-stationarity (Costantini and Martini, 2010). In the current study, two tests—the Augmented Dickey-Fuller (ADF) and the Phillips Parron (PP)—were used, as suggested by Agbenyo et al. (2022), to test for unit root problems. Although both tests serve the same general objective, the PP test has the additional benefit of detecting unidentified autocorrelation concerning a structural break in the estimation model (Addey, 2019). All the variables were found to be stationary at level or first differencing, which satisfied the ARDL model's prerequisite for its application (Table 2A).

2.4.2. Multi-collinearity testing

When dealing with multiple variables, as in the current study, there are possibilities of correlation among data points causing multi-collinearity (Gujarati, 2022). If multi-collinearity is present, it causes model overfitting, which leads to false conclusions. The study conducted variance inflation factor (VIF) and tolerance level tests and observed both these values to remain within the permissible limits (VIF$<$10; tolerance level $>$ 0.1), indicating no multicollinearity (Table 2B; Gujarati, 2022).

2.4.3. Ideal lag length calculation criteria

In econometric modeling, as in ARDL, the impact on a response variable (regressand) by an explanatory variable (regressor), *i.e.*, climatic parameters, rarely occurs all at once, but instead happens over time (De Boef and Keele, 2008). The same applies to the crop-climate link, which exhibits a residual effect that is spread over years, and the ARDL model uses lagged (previous year) values of variables to regress against the current year dependent variable to capture this effect by determining how far back in time this effect is distributed (Warsame et al., 2021). The number of previous years that are significantly valuable to include in the model is referred to as an ideal lag length. Statistically, ideal lag length is one at which endogeneity and residual correlation are at their lowest and model power is at its highest (Nkoro and Uko, 2016). As per the literature, the statistical tests prevalently used for this purpose are likelihood ratio (LR), final prediction error criterion (FPE), Akaike information criterion (AIC), Schwarz Bayesian information criterion (SC), and Hannan-Quinn information criterion (HQ) (Pickson et al., 2020). The ideal lag order was discovered to be "one" based on the minimum values produced by all tests (Table 2C), meaning that it is worthwhile to regress previous one-year values of variables against the response variable in the ARDL model.

2.4.4. Cointegration testing procedure

Before beginning to generate the model coefficients, confirming a relationship between the independent and dependent variables is essential (Warsame et al., 2021). The Wald F-test was used, and its calculated value (4.84) was compared with the upper bound (3, 3.38, and 4.15) and lower bound critical values (2.08, 2.39, and 3.06) shown in Table 2D. The relationship exists if the F-calculated value is greater

Table 2
Results of the pre-analysis diagnostic tests that are required to perform before inputting the study variables to the ARDL model for impact assessment of climatic variables on the cotton yield.

(A) Unit root test results following Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests of variables including maximum temperature (Tmax), minimum temperature (Tmin), CO_2 emissions (COE), harvested area (HA), precipitation (PR), and cotton yield (Y)

Variables	ADF		PP	
	Level	First difference	Level	First difference
Tmax	−6.005***		−6.860***	
Tmin	−5.687***		−11.121***	
COE	−2.256	−8.400	−2.264	−8.357***
HA	−2.696	−7.104	−2.705	−8.027***
PR	−6.415***		−6.544***	
Y	−5.807***		−5.807***	

"***" shows the significance level at 1%.

(B) Multicollinearity test results based on variance inflation factor (VIF) and tolerance value (TOV) tests of variables including maximum temperature (Tmax), minimum temperature (Tmin), CO_2 emissions (COE), harvested area (HA), and precipitation (PR)

Variable	Variance inflation factor (VIF)	Tolerance
Tmax	5.18	0.193072
Tmin	3.84	0.260679
COE	1.60	0.623611
PR	3.14	0.318447
HA	1.38	0.725874
Mean value	3.03	0.425558

(C) Model's lag selection criterion using sequential modified statistics test (SMLR), final prediction error (FPE) test, Akaike information criterion (AIC) method, Schwarz information criterion (SIC) method, and Hannan-Quinn information criterion (HQ) method

Lag	SMLR	FPE	AIC	SIC	HQ
0	NA	8.32e-13	−10.787	−10.548	−10.697
1	179.780*	4.02e-14*	−13.831*	−12.162*	−13.206*
2	46.287	5.13e-14	−13.669	−10.568	−12.507
3	45.441	5.68e-14	−13.786	−9.255	−12.089
4	30.698	1.01e-13	−13.683	−7.720	−11.449

* Indicates lag order selected by the criterion, SMLR: sequential modified likelihood ratio test statistic, FPE: Final prediction error, AIC: Akaike information criterion, SC: Schwarz information criterion, HQ: Hannan-Quinn information criterion, and each test at 5% level of significance.

(D) The ARDL bounds cointegration test results.

Test Statistic	Value	Significance (%)	Level I (0)	First difference I (1)
F-statistic	4.837	10	2.08	3
		5	2.39	3.38
		2.5	2.70	3.73
		1	3.06	4.15

than the upper determined values, doesn't exist if lower than lower bound values, and is inconclusive if between upper and lower bound values (Pickson et al., 2020). The calculated Wald-F values from the study were higher than the upper bound critical values, confirming a relationship between independent and dependent variables (Table 2D), and to start with further analysis for obtaining the ARDL coefficients to quantify the relationship between variables.

2.4.5. Post-analysis diagnostic tests, and sensitivity/robustness check of ARDL model

The next procedure was to assess the fitness of the residuals or error terms by testing the serial correlation and heteroskedasticity after estimating the elasticity coefficients of the final fitted regression equation of the ARDL model (Ding and Shi, 2022). Serial correlation occurs when the error terms auto-correlate with their lag values, causing biases in standard errors and lead to incorrect F-test or *t*-test applications in drawing conclusions because of the higher possibility of type-1 or type-2 errors (Coad, 2007). The model was found to be free of the autocorrelation issue according to the BGLM test findings (Table 3A). The time-series or any multiple regression model assumes that the residuals/error terms should be evenly scattered and have constant variance, referred to as homoskedasticity (Astivia and Zumbo, 2019). Violation of this assumption is termed heteroskedasticity, which must be tested before making inferences on the model output. The current study used the Breusch–Pagan–Godfrey (BPG) test, and the results confirmed no heteroskedasticity in the residuals of the model output (Table 3A). Another crucial factor is parameter constancy, which provides insight into whether the output coefficients are sufficiently stable to generate reliable inferences over the study period (Deng and Perron, 2008). Two tests—the cumulative sum (CUSUM) test and the CUSUM of squares

Table 3
Post-analysis diagnostic testing.

(A) Diagnostic test results following Breusch–Pagan–Godfrey (BPG) test, Breusch-Godfrey Lagrange Multiplier (BGLM) test, cumulative sum (CUSUM) and CUSUM of squares of recursive residuals tests.

Test	Statistics	Probability
BPG test for Heteroskedasticity	1.323	0.260
BGLM test for Serial Correlation	1.014	0.372
CUSUM	Stable (Fig. 3)	
CUSUM Squares	Stable (Fig. 4)	

(B) Fully modified ordinary least square (FMOLS) model results for robustness revalidation of the study model.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Tmax	−5.672	2.496	−2.272	0.028
Tmin	3.109	1.358	2.290	0.027
COE	0.680	0.220	3.092	0.004
PR	−0.266	0.242	−1.102	0.277
C	12.209	7.002	1.744	0.088
R-Square	0.620			
Adjusted R-Square	0.576			

Tmax is maximum temperature, Tmin is minimum temperature, PR is precipitation, and COE is CO_2 emissions.

test—were carried out for parameter constancy. Systemic breaks (intercept terms) from the stability of the model coefficients are measured by the CUSUM test, and rapid/abrupt breaks (residual variance) are measured by the CUSUM of squares (Hansen, 1991). The consistency, stability, and persistence of parameter plot lines within the critical bounds at a 5 % significance level revealed the output coefficients of the ARDL to be accurate and stable throughout the study period (Fig. 3; Fig. 4).

3. Results and discussions

3.1. Climatic trend analysis (Yr. 1970 to 2020)

The present study noted that the Tmax, Tmin, and average temperature (Tavg) throughout the CGP were computed to be 29.95 °C, 17.55 °C, and 23.75 °C during 1970–2020 in MS (Table 4). These computed values of MS's Tmax, Tmin, and Tavg were found more than those (29.42 °C, 17.03 °C, and 23.22 °C) recorded in the southeastern US, by 0.53 °C, 0.52 °C, and 0.53 °C, respectively. Notably, this differential amount itself is comparable to the magnitude of warming recorded across the southeastern US (Sharma et al., 2022b). The DTR and PR were measured at 12.41 °C and 241.60 mm, respectively, in MS (Table 4).

The Mann-Kendall estimates revealed that the Tmax in MS during CGP followed an insignificant (p-value of 0.512) positive trend from 1970 to 2020 (Fig. 5A). These results were corroborated by Shammi and Meng (2021) in context of MS, and by Sharma et al. (2022a) in context of the US's southeastern region. In MS, the Tmin displayed a significantly ($p < 0.001$) upward trend, the DTR exhibited a significantly ($p < 0.001$) downward trend, and the PR showed non-significant (p-value of 0.373) trend (Fig. 5B, 5C, 5D). Regarding PR, the current results agree with Sharma et al. (2022a) who reported weak PR trends in the southeastern US. Furthermore, Tmin findings were similar as noted by Shammi and Meng (2021); their study was focused on the soybean growing season as no such estimates are available so far for CGP in MS. For MS or the southeast US, the DTR trend estimation is still a new parameter; hence, there are currently no prior estimates with which to compare the current findings. However, the asymmetric warming caused by different rates of diurnal and nocturnal warming is attributed in the literature to the decreasing trend of DTR, which is recognized to be more concerning to plant physiological processes (Doan et al., 2022).

Sen-slope analysis on quantifying the climatic trend in MS revealed that Tmax during CGP had risen by 0.30 °C from 1970 to 2020 with an annual rate of 0.006 °C, or 0.06 °C decade^{-1} (Table 4). The Tmin during CGP, on the other hand, showed a warming by 1.17 °C between 1970 and 2020, with a rate of 0.23 °C decade^{-1} or 0.023 °C per year (Table 4). To determine the speed at which the ecosystems must readjust, it is essential to quantify the climate trend (Smith et al., 2015). These Tmin changes are typically thought to be more influential than that of Tmax in affecting crop reproductive processes and photosynthetic activities. However, the adverse effects of the elevated Tmax on crops are further exacerbated when drought and a high number of sunshine hours coexist (Parent and Tardieu, 2014). Further investigation after the Sen-slope calculations reveals that the shift in Tmin (1.17 °C) had greater implications on warming the Tavg than the shift in Tmax (0.30 °C) (Table 4). This showed that Tmax had contributed 20.4 % whereas Tmin had contributed 79.6 % to the overall warming over the research period in MS, indicating that Tmin accounted for most of the warming. Previously, Karl et al. (1993) and most recently Doan et al. (2022) also reported higher contributions of Tmin than that of Tmax in global climate warming. Between 1970 and 2020, the DTR and PR changed by −1.07 °C and 22.54 mm, respectively, during CGP (Table 4). For DTR, the rate of change was reported to be −0.021 °C year^{-1} or −0.21 °C decade^{-1}, and for PR, 0.442 mm year^{-1} or 4.42 mm decade^{-1} (Table 4). Globally, Sun et al. (2019) and within the US, Qu et al. (2014) noted similar results in terms of DTR. Similarly, PR rates noted in current study are comparable to other studies in the southeastern US focused on different crop-growing seasons (Sharma et al., 2022a; Sharma et al., 2022b).

3.2. Estimates of climate change impact on cotton yield

The long- and short-term estimations for the effects of change in Tmax, Tmin, COE, and PR from 1970 to 2020 under CGP on MS cotton yield, are presented in Table 5. The long-term ARDL coefficient for Tmax was found to be negative and significant (p-value of 0.012), indicating a decrease in cotton yield due to the ongoing Tmax trend in MS (Table 5), which are in line with the recent findings of Eck et al. (2020) and Sharma et al. (2022b). However, the short-term coefficient was discovered to be insignificant and negative (Table 5), indicating no effect.

The reasoning for the short-term effect's insignificance can be better understood by revisiting the interpretation of the econometric model

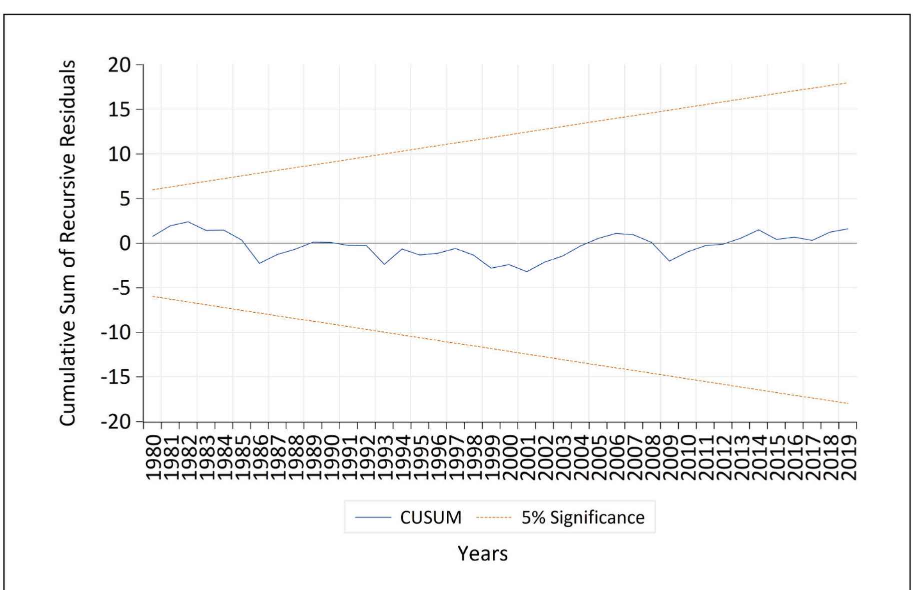

Fig. 3. A plot of cumulative sum (CUSUM) of recursive residuals showing stability of the ARDL model's coefficients.

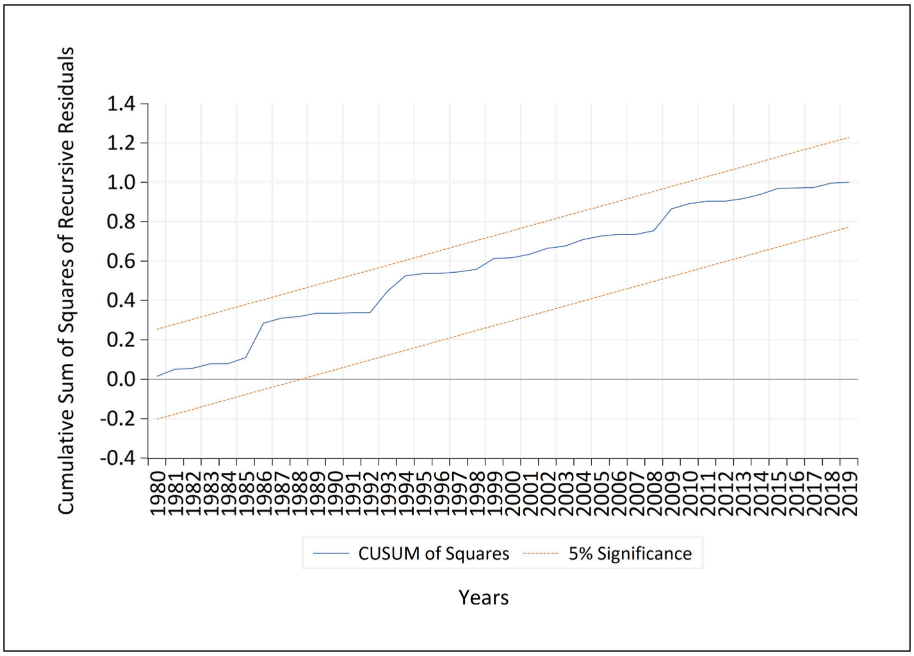

Fig. 4. A plot of cumulative sum (CUSUM) of squares of recursive residuals showing stability of the ARDL model's coefficients.

Table 4
The summarized results of the Mann-Kendall test (tau) and the Sen's slope (coefficient) method for trend estimation of study variables in Mississippi from 1970 to 2020.

Particulars	Mean	Z_k	Sen-slope	Change in climatic variable (1970–2020)
Cotton growing period				
Tmax (°C)	29.95 (0.753)	0.063	0.006	0.30
Tmin (°C)	17.55 (0.693)	0.338^{***}	0.023	1.17
DTR (°C)	12.41 (0.723)	-0.312^{***}	−0.021	−1.07
PR (mm)	241.60 (48.871)	0.086	0.442	22.54

Zk is Kendall's test standardized value. Zk negative (−) value signifies a downward (decreasing) trend, and a positive (+) value indicates an upward (increasing) trend. The Sen slope value represents the rate of change per year. The negative (−) value of Sen slope means the rate of decrease per year, while the positive (+) value represents the rate of increase per year. Significance: "*" $p < 0.05$, "**" $p < 0.01$, and "***" $p < 0.001$. Tmax is maximum temperature, Tmin is minimum temperature, DTR is diurnal temperature range, and PR is precipitation.

presented in section 2.3 (study model), which states that the Tmax trend's immediate impact in the short run was too meager to be noticeable at once, but over time, the cumulative effect became noticeable to significantly impact cotton yield in the long run. More specifically, every 1 °C Tmax rise during CGP significantly decreased the cotton yield by 6.10 % in the long run (Table 5). These results are consistent with previous findings from econometric studies (Diarra et al., 2017; Raju et al., 2014), simulation modeling studies (Ton, 2011; Hebbar et al., 2013; Reddy et al., 2002; Li et al., 2022), *meta*-analysis studies (Li et al., 2021), and review studies (Reddy et al., 1997). With every 1 °C Tmax warming, literature has reported yield reductions of 2.01 % (Arshad et al., 2021), 6.25 % (Schlenker and Roberts, 2009), 10 % (Pettigrew, 2008), and even up to 34 % (Sharma et al., 2022b). Yet, other studies have provided contradictory findings, such as beneficial effects of Tmax warming on cotton yield (Chen et al., 2015; Abbas, 2020, Li et al., 2020). The contradiction in the literature is due to several reasons, including the diversity of geographical settings with varying rates, directions, magnitudes, and intensifications of warming, as well as the use of crop simulation or statistical models. Moreover, this disagreement could be understood via two studies by Hedhly et al. (2009) and Reddy et al. (1992a). According to Hedhly et al. (2009), whether the prevalent Tmax is lower than or relatively close to the ideal temperature determines if any additional Tmax warming will positively or negatively impact yield. Reddy et al. (1992a) deduced that the ideal cotton temperature is 28 °C. Above 30 °C, the vegetative and reproductive stages become unbalanced, but temperatures over 32 °C negatively impacts cotton yield. Nevertheless, the current study recorded a Tmax of 29.95 °C, which is already higher than the optimum (28 °C), but June (31.66 °C), July (32.93 °C), and August (32.78 °C) in MS were noted to be even higher than 30 °C or 32 °C (Table 4). Due to this, the cotton's reproductive phase (flowering and the initial stage of boll development) occurrence in MS was elevated above the optimal temperature range and close to the cardinal range, which ultimately had a detrimental effect on cotton yield, as shown in Table 5. Also, the photosynthetic mechanism is disrupted by this elevated Tmax, which also advances the phonologies (planting-to-anthesis and anthesis-to-maturity) (Abbas et al., 2019). In addition to these direct effects, this elevated Tmax also causes a reduction in water use efficiency (up to 60 %) due to the increased evaporative rates (Tao et al., 2006).

The ARDL coefficient for Tmin during CGP was determined to be positive and significant (p-value of 0.032) (Table 5), indicating that the Tmin ongoing trend in the MS had a positive effect on cotton yield in the long term. However, the short-term coefficient for Tmin was determined to be insignificant revealing no evidence of any improvement in cotton yield (Table 5). More specifically, an increase of cotton yield by 5.51 % was noted in the current study with every 1 °C rise in Tmin which is consistent with the findings of Schlenker and Roberts (2008), Chen et al. (2015), Eck et al. (2020), Sharma et al. (2022b), and Umer Arshad et al. (2022); all of which concluded that Tmin warming has a beneficial effect on cotton yield. Diarra et al. (2017) documented a 3 % yield increase with every 1 °C Tmin rise. Although there is no explicit agreement in the literature so far on the physiological response of plants (cotton) to Tmin (Song et al., 2022), a few studies reported that every 1 °C increase in Tmin expanded the growth stages, *i.e.*, 5-leaf stage-to-budding, budding-to-anthesis, and anthesis-to-ball opening resulting in yield improvement

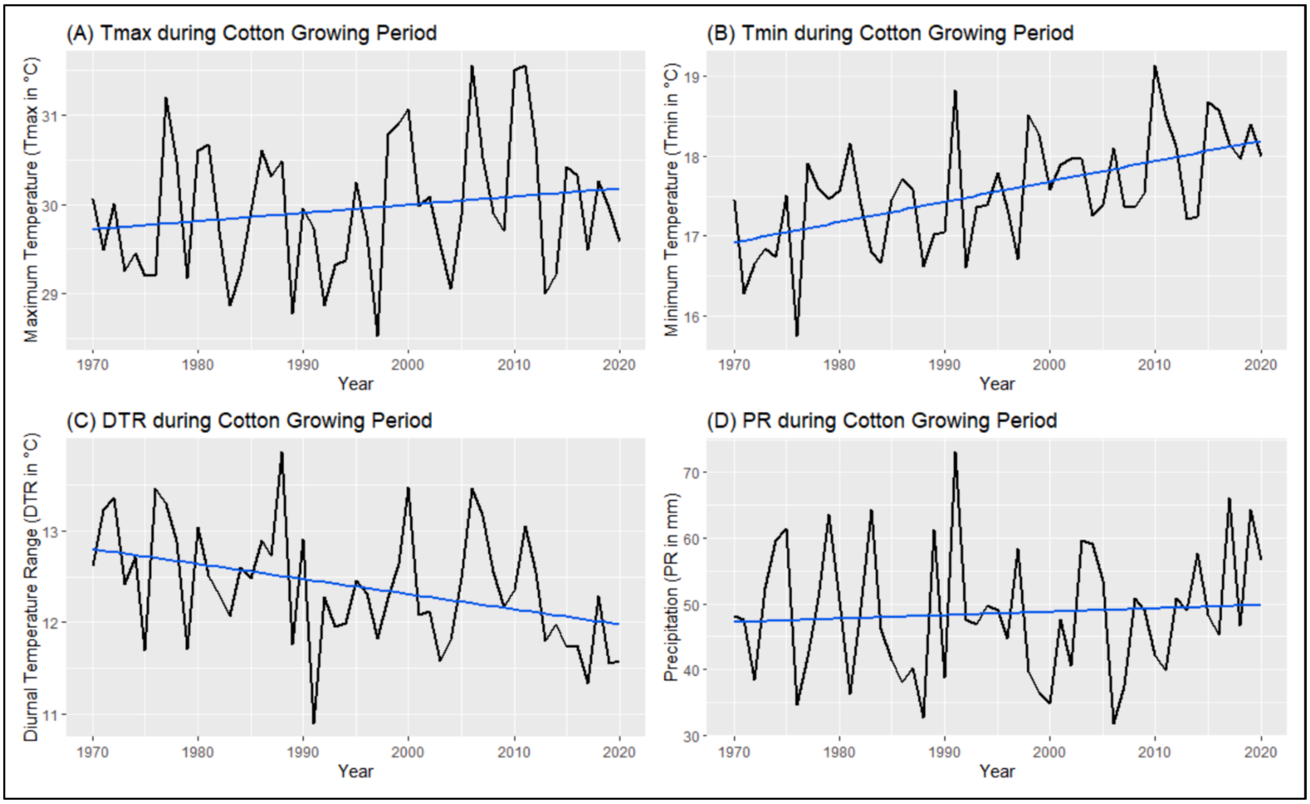

Fig. 5. Trend lines depicting the slope (positive or negative) of the trend for Mississippi (MS)'s cotton growing period (CGP) from 1970 to 2020 for (A) maximum temperature (Tmax), (B) minimum temperature (Tmin), (C) diurnal temperature range (DTR), and (D) precipitation (PR). The 'ggplot2' R package was used as the Source to create the figure (https://ggplot2.tidyverse.org/).

Table 5
Impact of climate change on cotton yield.

Calculated ARDL model estimates for short and long-run effects of Tmax, Tmin, COE, and PR on cotton yield (dependent variable)				
ARDL model long-run effects.				
Variable	Coefficient	Standard Error	t-Statistic	Probability
Tmax	−6.102	3.859	−1.581	0.012
Tmin	5.509	2.476	2.225	0.032
COE	0.453	0.349	1.298	0.020
PR	−0.434	0.378	−1.147	0.258
ARDL model short-run effects.				
Tmax	−3.123	2.466	−1.266	0.213
Tmin	1.485	1.388	1.069	0.292
COE	0.965	0.600	1.607	0.116
PR	−0.027	0.220	−0.124	0.901
C	8.174	10.686	0.765	0.449
ECM	−0.568	0.091	−6.240	0.000
R Square	0.726	–	–	–
Adjusted R Square	0.672	–	–	–

Tmax denotes maximum temperature, Tmin is minimum temperature, COE is CO_2 emissions, PR is precipitation, and ECM is error correction model.
The authors of this manuscript have no conflict interests to disclose.

(Wang et al., 2008). The Tmin (17.55 °C) reported in the current study falls in the range that was previously estimated to be beneficial for boll maturity (Table 4; Yfoulis and Fasoulas, 1978). There is evidence in the literature that the Tmin of 22 °C was discovered to be ideal for cotton in terms of maximizing the development of fruiting branches, expanding the stem and leaf area, accumulating biomass, and producing bolls (Reddy et al., 1992b). Hedhly et al. (2009) claimed that until Tmin does not exceed this range, its increasing trend will have a positive effect, by bringing the temperature closer to the optimal range and delaying the adverse impact of the rising trend on cotton production. Moreover, in the current study, Tmin was measured to be 17.55 °C; even the warmest months of June (19.61 °C), July (21.29 °C), and August (20.73 °C) have not crossed that threshold to harm cotton in MS during the study period (Table 4). In addition, it is worth noting that the Tavg plays a critical role in determining whether Tmin will have a favorable or unfavorable impact on cotton yield in the scenario where Tmax has already exceeded the optimal, as is the case with this study (Sharma et al., 2022b). In this case, if the rate of Tmin rise caused the Tavg to rise above 35 °C, it would slow down the participation of Rubisco by adversely affecting its activity in photosynthesis and respiration, ultimately reducing the yield (Loka and Oosterhuis, 2010; Sharwood, 2017). However, in MS, Tavg never exceeded this threshold throughout the study, and not even the three warmest months of June (25.63 °C), July (27.11 °C), and August (26.76 °C) (Table 4). This demonstrated that the ongoing Tmin warming rate (0.23 °C decade^{-1}) has not yet caused the Tavg to increase to the point where it can start negatively impacting cotton. Additionally, it was observed that the current study's Tavg in the months (June, July, and August), including the CGP Tavg, fell between the 20–30 °C range that was claimed to be ideal for cotton growth, making the Tmin effect on cotton yield favorable (Reddy et al., 1992a; Reddy et al., 1992b). The current study findings on Tmin (Table 5A), however, contradicts previous studies by Roussopoulos et al. (1998), Cetin and Basbag (2010), Loka and Oosterhuis (2010), Sankaranarayanan et al. (2010), Hebbar et al. (2013), Lokhande and Reddy (2014), and Echer et al. (2014) that found Tmin warming had a negative impact on cotton. Sawan (2018) noted no effect of Tmin warming on cotton. Further investigation revealed that the reason for the disagreement with these studies was that their Tmin values (>22 °C) were based on predictions made for the years 2050 or 2100, and as a result, they tested Tmin or Tavg values that were higher than those observed in the current study when running their estimation models. Furthermore, Li et al. (2021) predicted that the negative effect of Tmin warming will eventually be noticeable once it increases by 2 °C. Moreover, considering the ongoing Tmin warming rate of 0.23 °C decade^{-1} in MS, it (2 °C) will be reached by the year 2060.

The long-term COE coefficient for cotton yield was calculated to be positive and significant (p-value of 0.020). In contrast, the short-term coefficient was revealed to be non-significant (Table 5), indicating that the historical trends from 1970 to 2020 in COE have benefited the MS cotton yield in the long run with no effect in the short run. More specifically, every unit rise in the COE during the study timespan increased cotton yield by 0.45 % in the long run (Table 5). Li et al. (2021) arrived at similar conclusions, quantifying a 0.05 % increase in cotton yield with every unit rise in COE. Based on academic research, COE has the greatest influence on cotton production out of the four leading greenhouse gases linked to climate change (Sankaranarayanan et al., 2010). At the field level, cotton plants typically extract 11.34 tons ha^{-1} of COE to generate fiber, seed, protein, and other plant parts (Sankaranarayanan et al., 2010; Mujtaba et al., 2023). Notably, most of the COE-cotton research to date has been done in experimental settings, but there hasn't been much done to explore this nexus on a larger scale by incorporating actual data as documented in the current study thereby narrowing down the number of similar studies to compare the current findings with (Attavanich and McCarl, 2014; Feng et al., 2023). However, evidence from the literature shows that the increased atmospheric COE raises the CO_2 concentration at the intercellular level, and this intercellular-based CO_2 was noted to decrease the stomatal conductance and enhance the net photosynthetic rate, biomass production, and eventually the yield, as documented by experimental/greenhouse studies. These improved photosynthetic rates are attributable to the elevated COE gradient from the atmospheric to the CO_2-fixation site at the cellular level (Seneweera and Norton, 2011). In addition to being a substrate for photosynthetic activity, CO_2 acts as a competitive inhibitor for Rubisco in its oxygenation process, slowing down photorespiration; as a result, an increase in atmospheric CO_2 accelerates the rate of CO_2 uptake by leaves and canopies (Reddy and Hodges, 2000; Zhao et al., 2003). Also, the plant's ability to efficiently utilize resources *i.e.*, nutrients, light, and water was noted to improve by rise in CO_2 levels, but this improvement is much faster in C3 (cotton) plants than in C4 (Sankaranarayanan et al., 2010). This rapidity results from Rubisco's higher carbon saturation point in C3 plants than it does in C4 plants. Rubisco is a critical enzyme that controls photosynthesis reduction and photorespiratory oxidation in the carbon cycle (Qu et al., 2021). According to the research, the predicted trends in atmospheric COE may cause a diminishing positive effect on C4 crops by 2050 and 2100, but for C3 crops like cotton, such predictions are yet to be made (Leakey et al., 2009; von Caemmerer and Furbank, 2003). Therefore, more research in MS incorporating farmer's field level actual data (county-clustered) separated over different growth phases is needed to identify the scenarios during which this COE-yield association will continue to be positively cointegrated.

The ARDL coefficient for PR was calculated to be negative but non-significant for both the short and long term, indicating that the PR trends in MS from 1970 to 2020 did not affect the cotton yield (Table 5). Chen et al. (2011) deduced that this weak (non-significant) relationship between PR trends and yield, when examined over longer periods is not uncommon, particularly for cropping systems or regions that depend on assured irrigation systems. Most of the cropland in MS is also irrigated, with furrow irrigation and center-pivot systems being the two most common types (Kebede et al., 2014). According to the output of the ARDL model (Table 5A), the coefficient for the error correction terms (ECM) was discovered to be negative (−0.568) and statistically significant ($p < 0.001$). This result showed that 56.8 % of the model's deviation from equilibrium adjusts in the short term to attain long-term equilibrium, which means 56.8 % of the immediate effects of the climate change (on cotton yield) transfer (add-on) cumulatively to establish a stable foundation for the long-term effects. This rate (56.8 %) of adjustment of the model's deviation or immediate effects was noted to be equivalent to the findings of previous studies (Abbas, 2020; Rashid et al., 2020). Overall, the study model was able to explain 67 % of the variation in MS cotton yield caused by changes in the climatic variables used in the study (Table 5A), which is deemed sufficient for any model to be effective (Ray et al., 2015; Bekuma Abdisa et al., 2022).

4. Study Limitations

Certainly, this study has a few limitations, however, findings from this study make basis for future advancements. The limitations are:

1. Although the current study included most of the variables of crop-climate relationship studies (Chen et al., 2015; Diarra et al., 2017; dos Santos et al., 2022), the inclusion of additional data on yield-impacting variables, such as the rate of technological advancement, fertilizer use, pest management, sunlight hours, cloud coverage, vapor pressure deficit, irrigation rates, crop evapotranspiration, evaporative demand, and wind speeds, could produce more accurate estimates and allow the model to exhibit higher predictive efficiency by lowering its residual components. The current study, however, was limited to the study variables because the aforesaid data were unavailable during the needed (minimum consistent 30 years) timeframe specified for climatic studies.
2. The data used in the current study were daily values averaged over a month, which is the method most frequently used in crop-climate literature (Chen et al., 2015; Diarra et al., 2017; Sharma et al., 2023b). However, if consistent data (at least for 30 years) on an hourly scale were available, the study could reveal a more in-depth understanding of the crop-climate concept by clearly differentiating the climatic impact on vegetative and reproductive stages for MS.
3. Given that it is an econometric model, the ARDL model runs regression on the study variables at ceteris paribus, which implies that the other variables are held constant, and the interaction aspect is left unexplored. However, the interaction between diverse sets of factors might produce more realistic outcomes for farmers and other agricultural stakeholders.

5. Concluding remarks

Until now, the rate at which climate is changing and its impact on cotton were unknown in MS. Major shifts in climate parameters were noted during the studied time frame (1970–2020), revealing that the Tmax, Tmin, DTR, and PR have all risen by 0.30 °C, 1.17 °C, −1.07 °C, and 22.54 mm, respectively, with rate per decade noted as 0.06 °C, 0.23 °C, −0.21 °C, 4.42 mm, respectively. The warming trend in MS was found to be asymmetrical, with Tmax contributing 20.4 % and Tmin 79.6 % to the total warming. Every year, 56.8 % of the immediate (short-term) effects of the climate change (on cotton yield) were found to transfer (add-on) cumulatively to establish a stable foundation for the long-term effects, indicating that the issue of climate change is getting severe with time. Precipitation had no effect on cotton yield in the long and short-term. Similarly, no short-term effects were noted on cotton yield due to the Tmax, Tmin, and COE. However, cotton yield significantly decreased with rise in Tmax and increased with rise in Tmin and COE in the long-term. Overall, a 1 °C increase in Tmax reduced cotton yield by 6.10 %, a 1 °C increase in Tmin improved it by 5.50 %, and a unit increase in COE increased it by 0.45 % over the long run. Taken together, the crop-climate link marked a varied sensitivity for MS cotton towards short and long-term.

CRediT authorship contribution statement

Ramandeep Kumar Sharma: Writing – original draft, Validation, Methodology, Conceptualization. **Jagmandeep Dhillon:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Pushp Kumar:** Formal analysis, Data curation. **K Raja Reddy:** Writing – review & editing. **Vaughn Reed:** Writing – review & editing. **Darrin M. Dodds:** Writing – review & editing. **Krishna N. Reddy:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data sources along with the weblink are already mentioned in the methodology section

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